Transportation problem

Balanced → $D = S$

Unbalanced → $D \neq S$

		Destination			
		1	2	3	supply
source	A	2	3	4	10
	B	5	7	9	30
	C	10	2	5	60
Demand		20	30	50	100/100

		Destination			supply
		1	2	3	
source	A	2	3	4	10
	B	5	7	9	30
	C	10	2	5	60
Demand		20	25	45	90/100

If $D > S$ add dummy row with cost zero
If $D < S$ add dummy column with cost zero

Methods of solving Transportation problem:

Phase I: Finding the initial basic solution.

(i) Northwest corner cell method (NWCM)

(ii) Least cost cell method (LCM)

(iii) Vogel's approximation method (VAM method)

Phase II: Optimizing the Basic Feasible solution.

- UV method.

(i) MODI method (ii) stepping stone method.

Example: Find the initial basic feasible solution to the following transportation problem by (a) Northwest corner cell method (b) Least cost cell method and (c) VAM method.

		Destination					
		A	B	C	D	E	supply
source	1	5	1	8	7	5	15
	2	3	9	6	7	8	25
	3	4	2	7	6	5	42
	4	7	11	10	4	9	35
Demand		30	20	15	10	20	

Solution: Here,

① By NWCM

Destination

Source	A	B	C	D	E	F	Supply
1	[15] 5	1	8	7	5	0	15
2	[5] 3	[20] 9	6	7	8	0	~~25~~ 20
3	4	[10] 2	[15] 7	[10] 6	[7] 5	0	~~42~~ 32 ~~17~~ 7
4	7	11	10	4	[13] 9	[22] 0	~~35~~ 22
Demand	~~20~~ 5	~~30~~ 10	15	~~10~~	~~20~~ 13	22	95/117

Since Total Demand ≠ Total supply. It is called unbalanced transportation problem. So add dummy variation in destination and each element is zero cost value.

∴ No. of row + No. of column = occupied cell

∴ $4+6-1 = 9$

∴ $9 = 9$ which is equal no. of occupied cell.

So the solution is feasible. Hence, total transportation cost is

$$T.C. = (15\times5)+(5\times3)+(20\times9)+(10\times2)+(15\times7)+(10\times6)+(7\times5)+(13\times9)+(22\times0)$$

$$=$$

② By LCM

Destination

Source	A	B	C	D	E	F	Supply
1	5	1	8	7	5	[15] 0	15
2	[18] 3	9	6	7	8	[7] 0	~~25~~ 18
3	[2] 4	[30] 2	7	6	[10] 5	0	~~42~~ ~~12~~ 10
4	7	11	~~10~~	[10] 4	9	0	35
Demand	~~20~~ 2	~~30~~	15	~~10~~	~~30~~ 20	22 7	

Destination

Source	A	B	C	D	E	F	Supply
1	5	1	8	7	5	0 (15)	15
2	3 (18)	9	6	7	8	0 (7)	25 ~~18~~
3	4 (12)	2 (20)	7	6	5 (14)	0	42 ~~22~~ ~~14~~
4	7	11	10 (15)	4 (10)	9 (10)	0	35 ~~25~~ ~~15~~
Demand	20 ~~12~~	20	15	10	20 ~~10~~	22 ~~7~~	

$m+n-1 = 6+4-1 = 9$

$\therefore$ $m+n-1$ is equal to no. of occuppied cell. so, ~~its~~ solution is feasible. Hence, total transportation cost is

$TC = (8\times3)+(12\times4)+(20\times2)+(10\times5)+(10\times9)+(15\times0)+(7\times0)$

$= \ldots$

(ii) By VAM

	A	B	C	D	E	F	Supply	Cost difference				
1	5	1 (15)	8	7	5	0	15	1	4	—	—	—	—
2	3 (20)	9	6	7	8	0	25 ~~5~~	2	3	3	3	1	2
3	4	2 (15)	7 (7)	6	5 (20)	0	42 ~~27~~ ~~2~~	2	2	2	1	1	2
4	7	10	10	4 (10)	9	0 (22)	35 ~~18~~ ~~3~~	4	3	3	3	5	1
Demand	20	20 ~~15~~	15 ~~8~~	10	20	22							
Cost difference	1	1	1	2	0	0							
	1	1	1	2	0	—							
	1	7	1	2	0	—							
	1	—	1	2	3	—							
	—	—	1	2	3	—							
	—	—	1	—	3								

c_{ij} = cost of transporting one unit of the product from i^{th} origin to j^{th} destination.

x_{ij} = quality transported from i^{th} origin to j^{th} destination.

Example: suggest an optimal transportation plan with a view to minimize cost and determine the minimum transportation cost from the following

Warehouse	Stores (cost per unit)				supply
	S_1	S_2	S_3	S_4	
W_1	9	12	9	6	7
W_2	7	3	7	7	6
W_3	6	5	9	11	9
Demand	6	4	6	2	

Solution: Here,

since $D \neq S$, to make the problem balanced, we need to dummy store for exceeds units 4 with per unit transportation cost is zero.

	S_1	S_2	S_3	S_4	S_5	supply	cost difference I	II	III	IV
W_1	~~9~~	~~12~~	[1] 9	[2] 6	[4] 0	~~7~~ 3	~~6~~	3	3	0
W_2	~~7~~	[4] 3	[2] 7	~~7~~	~~0~~	~~6~~ 2	3	4	0	0
W_3	[6] 6	~~5~~	[3] 9	~~11~~	~~0~~	~~9~~ 8	5	1	3	3
Demand	~~6~~	~~4~~	~~6~~ 4 3	~~2~~	~~4~~	22				
Cost difference I	1	2	2	1	0					
II	1	2	2	1	—					
III	1	—	2	1	—					
IV	1	—	2	—	—					

Initial transportation cost $= 1\times9 + 2\times6 + 4\times0 + 4\times3 + 2\times7 + 6\times6 + 3\times9 =$ Rs. 110

Test of degeneracy:

$m+n-1 = 3+5-1 = 7 =$ No. of occupied cells. It is non-degeneracy.

Testing of optimaly condition (use MODI):

Warehouse	stores (cost per unit in Rs)					supply	R_i
	S_1	S_2	S_3	S_4	S_5		
W_1	9	12	[1] 9	[2] 6	[4] 0		$R_1 = 0$
W_2	7	[4] 3	[2] 7	7	0		$R_2 = -2$
W_3	[6] 6	5	[3] 9	11	0		$R_3 = 0$
Demand	6	4	6	2	4		
	$K_1 = 6$	$K_2 = 5$	$K_3 = 9$	$K_4 = 6$	$K_5 = 0$		

calculation of raw and column values for occupied cell. By using $c_{ij} = R_i + K_j$ and most occupied cell in R_1 $\therefore R_1 = 0$.

$c_{15} = R_1 + K_5 \Rightarrow 0 = 0 + K_5 \Rightarrow K_5 = 0$

$c_{14} = R_1 + K_4 \Rightarrow 6 = 0 + K_4 \Rightarrow K_4 = 6$

$c_{13} = R_1 + K_3 \Rightarrow 9 = 0 + K_3 \Rightarrow K_3 = 9$

$c_{23} = R_2 + K_3 \Rightarrow 7 = R_2 + 9 \Rightarrow R_2 = -2$

$c_{22} = R_2 + K_2 \Rightarrow 3 = -2 + K_2 \Rightarrow K_2 = 5$

$c_{31} = R_3 + K_1 \Rightarrow 6 = 0 + K_1 \Rightarrow K_1 = 6$

$c_{33} = R_3 + K_3 \Rightarrow 9 = R_3 + 9 \Rightarrow R_3 = 0$

calculation of improvement indices for unoccupied cells using the relation $d_{ij} = c_{ij} - (R_i + K_j)$

$d_{11} = 9 - (0+6) = 3$

$d_{12} = 12 - (0+5) = 7$

$d_{21} = 7 - (-2+6) = 3$

$d_{24} = 7 - (-2+6) = 3$

$d_{25} = 0 - (-2+0) = 2$

$d_{32} = 5 - (0+5) = 0$

$d_{34} = 11 - (0+6) = 5$

$d_{35} = 0 - (0+0) = 0$

since, $d_{ij} \geq 0$. optimal solution is obtained. Hence optimal transportation cost = Rs. 110.

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Question: Apply MODI method to obtain optimum solution of transportation problem using the data.

Source \ Destination	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

Solution:

Source \ Destination	D_1	D_2	D_3	D_4	Supply	cost difference I	II	III	IV
S_1	[5] × 19	× 30	× 50	[2] × 10	~~7~~ 2	9	9	40	40
S_2	× 70	× 30	[7] 40	[2] 60	~~9~~ 2	10	20	20	20
S_3	× 40	[8] × 8	× 70	[10] × 20	~~18~~ 10	12	20	← 50	—
Demand	~~5~~	~~8~~	~~7~~	~~14~~ 4 2					

cost difference				
I	21	22 ↑	10	10
II	21 ↑	—	10	10
III	—	—	10	10
IV	—	—	10	50 ↑

$m = 3$, $n = 4$, No. of occupied cell $= 6$

$\therefore m+n-1 =$ No. of occupied cell.

Hence, initial solution using VAM method is

$$\text{Total cost} = 5\times19 + 2\times10 + 7\times40 + 2\times60 + 8\times8 + 10\times20$$
$$= \text{Rs. } 779$$

Therefore, $m+n-1 =$ No. of occupied cell, then initial basic ~~solution~~ feasible solution and allocation is independent. So optimally solution is satisfied.

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calculation of u_i and v_j in relation $c_{ij} = u_i + v_j$ for occupied cell.

Source \ Destination	D_1	D_2	D_3	D_4	Supply	u_i
S_1	19 [5]	30	50	10 [2]	7	$u_1 = 10$
S_2	70	30	40 [7]	60 [2]	9	$u_2 = 60$
S_3	40	8 [8]	70	90 [10]	18	$u_3 = 20$
Demand	5	8 [8]	7	7		
v_j	$v_1 = 9$	$v_2 = -12$	$v_3 = -20$	$v_4 = 0$	~~$v_5 =$~~	

~~$c_{11} = u_1 + v_1 \Rightarrow 19 = 10 + v_1 \therefore v_1 = 9$~~

$\therefore$ Maximum occupied cell in D_4. So $v_4 = 0$.

$c_{14} = u_1 + v_4 \Rightarrow 10 = u_1 + v_4 \Rightarrow 10 = u_1 + 0 \therefore u_1 = 10$

$c_{24} = u_2 + v_4 \Rightarrow 60 = u_2 + 0 \therefore u_2 = 60$

$c_{34} = u_3 + v_4 \Rightarrow 20 = u_3 + 0 \therefore u_3 = 20$

$c_{11} = u_1 + v_1 \Rightarrow 19 = 10 + v_1 \therefore v_1 = 9$

$c_{23} = u_2 + v_3 \Rightarrow 40 = 60 + v_3 \therefore v_3 = -20$

$c_{32} = u_3 + v_2 \Rightarrow 8 = 20 + v_2 \therefore v_2 = -12$

calculation of d_{ij} in relation $d_{ij} = c_{ij} - (u_i + v_j)$ in un-occupied cell is

$d_{12} = c_{12} - (u_1 + v_2) = 30 - \{10 + (-12)\} = 32$

$d_{13} = c_{13} - (u_1 + v_3) = 50 - \{10 + (-20)\} = 60$

$d_{21} = c_{21} - (u_2 + v_1) = 70 - (60 + 9) = 1$

$d_{22} = c_{22} - (u_2 + v_2) = 30 - \{60 + (-12)\} = -18$

$d_{31} = c_{13} - (u_3 + v_3) = 40 - (20 + 9) = 11$

$d_{33} = c_{33} - (u_3 + v_3) = 70 - \{20 + (-20)\} = 70$

~~$d_{ij} < 0$, i.e.~~

<u>optimally test</u>:

$d_{ij} < 0$, i.e. $d_{22} = -18$ (most negative value). So x_{22} is entering the basis.

construction of loop and allocation of unknown quantity x.

$\therefore x = \text{minimum of } (8, 2) = 2$

Destination / Source	D_1	D_2	D_3	D_4	Supply	u_i
S_1	19 [5]	30	50	10 [2]	7	$u_1 =$
S_2	70	30 $+x$	40 [7]	60 $2-x$	9	$u_2 =$
S_3	40	8 [$8-x$]	70	20 [$10+x$]	18	$u_3 =$
Demand	5	8	7	14		
v_j	$v_1 =$	$v_2 =$	$v_3 =$	$v_4 =$		

Then,

Modified solution is given

Destination / Source	D_1	D_2	D_3	D_4	Supply	u_i
S_1	19 [5]	30	50	10 [2]	7	$u_1 = -10$
S_2	70	30 [2]	40 [7]	60	9	$u_2 = 22$
S_3	40	8 [6]	70	20 [12]	18	$u_3 = 0$
Demand	5	8	7	14		
v_j	$v_1 = 29$	$v_2 = 8$	$v_3 = 18$	$v_4 = 20$		

$\therefore m+n-1 =$ No. of occupied cells, then

$$Tc = 5\times19 + 2\times10 + 2\times30 + 7\times40 + 6\times8 + 12\times12$$
$$= \text{Rs. } 743$$

since, $m+n-1 =$ No. of occupied cells. It is non-degeneracy and independent.

calculation of u_i and v_j in occupied cell by the relation

$c_{ij} = u_i + v_j$.

~~$c_{11} = u_1 + v_1$~~ We choose most occupied cell, i.e. $u_3 = 0$.

$\therefore c_{32} = u_3 + v_2 \Rightarrow 8 = 0 + v_2 \therefore v_2 = 8$

$c_{34} = u_3 + v_4 \Rightarrow 20 = 0 + v_4 \therefore v_4 = 20$

$c_{14} = u_1 + v_4 \Rightarrow 10 = u_1 + 20 \therefore u_1 = -10$

$c_{22} = u_2 + v_2 \Rightarrow 30 = u_2 + 8 \therefore u_2 = 22$

$c_{11} = u_1 + v_1 \Rightarrow 19 = -10 + v_1 \therefore v_1 = 29$

$c_{23} = u_2 + v_3 \Rightarrow 40 = 22 + v_3 \therefore v_3 = 18$

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calculation of d_{ij} by the relation $d_{ij} = c_{ij} - (u_i + v_j)$ in un-occupied cell

$d_{12} = c_{12} - (u_1 + v_2) = 30 - (-10 + 8) = 32$

$d_{13} = c_{13} - (u_1 + v_3) = 50 - (-10 + 18) = 42$

$d_{21} = c_{21} - (u_2 + v_1) = 70 - (29 + 22) = 19$

$d_{24} = c_{24} - (u_2 + v_4) = 60 - (22 + 20) = 18$

$d_{31} = c_{31} - (u_3 + v_1) = 40 - (0 + 29) = 11$

$d_{33} = c_{33} - (u_3 + v_3) = 70 - (0 + 18) = 52$

<u>optimally Test:</u>

$d_{ij} \geq 0$, then optimally solution is obtained with minimum transportation cost is Rs. 743 <u>Ans.</u>

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